Technical Justification Note

Why V2 Used a Hybrid 3D-Conv + ResNet-18 Visual Encoder Instead of a Plain 3-Block 3D-CNN

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1 Motivation and Problem Statement

Our initial visual encoder (**V1**, `conv3d_bilstm` in `lipscribe/models/visual_encoder.py`) implemented three plain 3D convolutional blocks (1→32→64→96 channels) followed by flattening the 96×6×6 = 3,456 spatial feature map into a dense `Linear(3456, 512)` layer and a 2-layer BiLSTM. While V1 converged on the training set and achieved 23.04 % overall test WER in standalone **B2 v1**, speaker-level diagnostics exposed a critical generalisation bottleneck:

- **2.5× inter-speaker error spread on unseen speakers:** Test WER ranged from 13.3 % on speaker `s24` to 33.7 % on speaker `s9`, with validation speakers averaging 35.74 % WER.
- **Coordinate-tied flatten bottleneck:** Nearly one-third of V1's parameters (1.77 M of 5.25 M) reside in a single `Linear(3456, 512)` matrix that ties weights to rigid (x, y) coordinates on the 6×6 grid, making embeddings sensitive to speaker lip morphology, jaw width, illumination, and minor MediaPipe crop shifts.

To provide speaker-invariant visual representations that align cleanly with our audio encoder in cross-modal attention, **V2** (`resnet18_bilstm`) replaces the shallow 3-block 3D-CNN with the canonical deep lipreading architecture (Stafylakis & Tzimiropoulos, 2017; Martinez et al., 2020): **per-clip photometric normalisation + a 5-frame 3D spatiotemporal stem + a 16-layer 2D ResNet-18 residual trunk + global spatial average pooling + a 2-layer BiLSTM**.

2 Architectural Comparison: V1 (`conv3d_bilstm`) vs. V2 (`resnet18_bilstm`)

3 Controlled Empirical Verification in Multimodal Fusion (`P3` vs. `P4`)

In standalone visual-only training, **B2 v2** was capped at a 50-minute time budget (**13 epochs** with heavy cutout and time-masking augmentation) compared with **28 epochs** for **B2 v1**, resulting in a higher standalone WER (32.20 % vs. 23.04 %). However, to isolate the true representational quality of the visual encoder for multimodal fusion, we ran a strictly controlled head-to-head experiment in **Proposed Model P**:

- **Model P3 (`p_init_b1_b2`):** Initialised from `B1` (audio) + **B2 v1 (`conv3d_bilstm`)**.
- **Model P4 (`p_init_b1_b2_v2`):** Initialised from `B1` (audio) + **B2 v2 (`resnet18_bilstm`)**.
- **Controlled protocol:** Identical 30-minute full-corpus training budget, identical cross-modal attention and gated fusion architecture, identical optimiser (lr = 3×10^{-4}), identical modality

Table 1: Structural comparison of the V1 and V2 visual encoders (`[B, 1, T, 96, 96] -> [B, T, 512]`).

Component	**V1:** `conv3d_bilstm`	**V2:** `resnet18_bilstm`	**Technical Benefit of V2**
Input Norm	Corpus stats (`none`)	Per-clip zero-mean, unit-var (`clip`)	Removes per-speaker brightness and contrast bias
3D Stem	3 stacked `Conv3d` blocks (3×5×5, 3×5×5, 3×3×3)	Single `Conv3d` (5×7×7, stride 1×2×2) + `MaxPool3d`	Captures 200 ms (5-frame) viseme transitions at input without 3D FLOP explosion
Spatial Trunk	None (only 3 conv layers total; no skip connections)	16-layer 2D ResNet-18 (8 `BasicBlock`s, 64→512 ch)	Deep hierarchical spatial features (lip contour, teeth, tongue) without vanishing gradients
Spatial Pooling	Flatten 96×6×6 = 3,456 → `Linear(3456, 512)`	`AdaptiveAvgPool2d(1)` over 3×3×512 map → 512-d	Enforces spatial translation invariance; eliminates dense flatten overfitting
Regularisation	Feature dropout (0.3), crop jitter (±4 px), flip	`Dropout2d(0.2)`, scale jitter (±10 %), cutout, time masks	Prevents memorisation of static speaker face geometry

dropout ($p_a = 0.25, p_v = 0.15$), and identical random seed (seed = 1).

4 Conclusion

Table 2 provides decisive empirical justification for the V2 `ResNet-18` front end:

1. **State-of-the-art fusion accuracy:** Even though `B2 v2` was pre-trained for only 13 epochs, its residual, clip-normalised, globally pooled features enable **Model P4** to cut clean test WER more than in half (2.13 % → 1.02 % greedy; 1.25 % → 0.71 % grammar) and reduce −5 dB WER by over a quarter (7.30 % → 5.34 %) relative to **Model P3** (`Conv3D`).
2. **2.25× stronger dynamic visual gating under noise:** In P3 (`Conv3D`), the fusion gate $\bar{g}$ remains heavily biased toward audio (0.884 → 0.834) because the shallow 3D-CNN features are noisier across unseen speakers. In P4 (`ResNet-18`), the gate actively trusts the visual stream, assigning 19.6 % weight to lip features in clean audio and shifting to **30.9 % visual weight at −5 dB** ($\bar{g} = 0.691$).

Table 2: Controlled full-corpus comparison: Model P3 (`Conv3D`) vs. Model P4 (`ResNet-18`) on held-out test speakers.

Evaluation Metric	**Model P3 (`Conv3D`)**	**Model P4 (`ResNet-18`)**	**Relative Gain of V2**
Best Mean Validation WER / CER	14.17 % / 8.53 %	**8.82 % / 5.17 %**	−37.8 % WER / −39.4 % CER
Test WER – Clean (Greedy / Grammar)	2.13 % / 1.25 %	**1.02 % / 0.71 %**	**−52.1 %** Greedy / **−43.2 %** Grammar
Test WER – 10 dB (Greedy / Grammar)	2.67 % / 1.62 %	**1.53 % / 1.10 %**	−42.7 % Greedy / −32.1 % Grammar
Test WER – 5 dB (Greedy / Grammar)	3.27 % / 2.08 %	**1.87 % / 1.42 %**	−42.8 % Greedy / −31.7 % Grammar
Test WER – 0 dB (Greedy / Grammar)	4.33 % / 2.88 %	**2.79 % / 2.06 %**	−35.6 % Greedy / −28.5 % Grammar
Test WER – −5 dB (Greedy / Grammar)	7.30 % / 5.20 %	**5.34 % / 3.95 %**	**−26.8 %** Greedy / **−24.0 %** Grammar
Mean Test WER Across All 5 SNRs	3.94 % / 2.61 %	**2.51 % / 1.85 %**	**−36.3 %** Greedy / **−29.1 %** Grammar
No-Visual Test Fallback (`--drop-mod visual`, Clean)	2.72 %	**1.88 %**	−30.9 % WER
Mean Fusion Gate Shift $\Delta\bar{g}$ (Clean → −5 dB)	0.884 → 0.834 (−0.050)	0.804 → 0.691 (−0.113)	**2.25× stronger visual gating**